CURRICULUM VITAE

Dr. Hoang Nguyen

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Profile

I am a security and privacy researcher with a doctorate (Dr. sc.) from the Institute of Information Security at ETH Zurich. I develop formal methods for building software systems that are secure and privacy-compliant, with a focus on access control, privacy by design, and data protection. My work is published at leading security and privacy venues, including papers at ACM CCS and PETS.

Education

Doctor of Science (Dr. sc.), Computer Science 2022 – 2026
ETH Zurich, Institute of Information Security Switzerland

Master of Science, Formal Methods in Computer Science 2020 – 2021
Universidad Autónoma | Complutense | Politécnica de Madrid Spain

Bachelor of Science, Computer Science 2014 – 2018
Vietnamese-German University | Frankfurt University of Applied Sciences Vietnam | Germany

Employment

Research and Teaching Assistant Sep 2026 – Present
Vietnamese-German University, Faculty of Engineering Vietnam

Doctoral Assistant Jul 2022 – Aug 2026
ETH Zurich, Institute of Information Security Switzerland

Research Intern Feb 2022 – Jun 2022
IMDEA Software Institute Spain

Research Assistant Aug 2021 – Jan 2022
DAAD German Academic Exchange Service Vietnam

Technical Research Assistant Feb 2019 – Aug 2020
Vietnamese-German University, Faculty of Engineering Vietnam

Software Engineer Feb 2018 – Jan 2019
Axon Active Vietnam

Awards and Scholarships

Vingroup Graduate Scholarship 2022 – 2026
Vingroup Science and Technology Scholarship Program Vietnam

DAAD Exchange Scholarship 2016
DAAD German Academic Exchange Service Germany

Merit Scholarship for Excellent Studies 2014 – 2018
Vietnamese-German University Vietnam

Skills

- **Security and privacy:** access control, data protection, privacy compliance
- **Formal methods and modeling:** model-driven engineering, UML/OCL, domain-specific languages, model transformations, code generation, logics and formal languages
- **Languages:** Vietnamese (native), English (professional), German (basic)

Research Output

Research Interest

My research lies at the intersection of security, privacy, and formal methods. I develop model-driven methodologies for building software systems that enforce security and privacy requirements by construction, with an emphasis on data protection: controlling who can access data, and ensuring that data is used only for the purposes for which it was collected.

Fine-Grained Access Control In the context of model-driven engineering, [**??**] discusses some of the challenges and opportunities of the mapping from the Object Constraint Language (OCL) to the Structured Query Language (SQL). [**??**] defines a novel mapping, called OCL2PSQL, that overcomes some of the aforementioned challenges. [**??**] proposes the definition of this OCL2PSQL mapping as a case to study the readiness of the state-of-the-art transformation tools. [**??**] formally characterizes the enforcement of Fine-Grained Access Control (FGAC) policies for a subset of SQL queries. [**??**] then proposes a model-based methodology for developing an FGAC enforcement mechanism following our formal characterization. [**??**, **??**, **??**, **??**] extend the methodology on several dimensions, namely, performance, expressiveness, and correctness.

Privacy and Data Protection We propose a model-driven development methodology that incorporates privacy policies into the system design [**??**]. In a nutshell, software engineers design a privacy model that supports notions of *purpose limitation*, *data subject consent* [**??**] and rely on predefined code generation to generate correct-by-design privacy enforcement [**??**]. We conduct a large-scale empirical study asking developers to implement privacy requirements to showcase the effectiveness of our approach compared to the code-centric development techniques [**??**, **??**].

Publications

1. **Hoàng Nguyễn**.
   Model-driven Privacy.
   Doctoral Thesis, ETH Zurich, 2026.

2. **Hoàng Nguyễn**, Srđan Krstić, and David Basin.
   Models Trump Code: An Empirical Analysis of GDPR Compliance.
   ACM Conference on Computer and Communications Security (CCS), 2026.

3. David Basin, François Hublet, Srđan Krstić, and **Hoàng Nguyễn**.
   Mechanizing Privacy by Design.
   ACM Conference on Computer and Communications Security (CCS), 2025.

4. Srđan Krstić, **Hoàng Nguyễn**, and David Basin.
   Model-driven Privacy.
   Privacy Enhancing Technologies Symposium (PETS), 2024.

5. David Basin, Juan Guarnizo, Srđan Krstić, **Hoàng Nguyễn**, and Martin Ochoa.
   Is Modeling Access Control Worth It?
   ACM Conference on Computer and Communications Security (CCS), 2023.

6. **Hoàng Nguyễn** and Manuel Clavel.
   An Extended Model-based Characterization of Fine-Grained Access Control for SQL Queries.
   Foundations Track, *Model Driven Engineering Languages and Systems* (MoDELS), 2023.

7. **Hoàng Nguyễn**.
   Intelligent Enforcement of Fine-Grained Access Control Policies for SQL Queries.
   Master's Thesis, Universidad Autónoma de Madrid, 2021.

8. **Hoàng Nguyễn**, Antonio García-Domínguez, and Manuel Clavel.
   The Transformation Tools Contest 2021: OCL2PSQL case.
   Software Technologies: Applications and Foundations (STAF), 3089:1–11, 2021.

9. **Hoàng Nguyễn** and Manuel Clavel.
   A Model-Driven Approach for Enforcing Fine-Grained Access Control for SQL Queries.
   SN Computer Science (SNCS), 2(5):370–389, 2021.

10. **Hoàng Nguyễn** and Manuel Clavel.
    Model-based Characterization of Fine-Grained Access Control Authorization for SQL Queries.
    Journal of Object Technology (JOT), 19(3):1–13, 2020.

11. **Hoàng Nguyễn** and Manuel Clavel.
    OCL2PSQL: An OCL-to-SQL Code-Generator for Model-Driven Engineering.
    Future Data and Security Engineering (FDSE), 11814:185–203, 2019.

12. Manuel Clavel and **Hoàng Nguyễn**.
    Mapping OCL into SQL: Challenges and Opportunities Ahead.
    OCL Workshop, *Model Driven Engineering Languages and Systems* (MoDELS), 2513:3–16, 2019.

13. **Hoàng Nguyễn**.
    On the Critical Configurations of the Chip Firing Game.
    Bachelor's Thesis, Vietnamese-German University, 2018.

Drafts

14. **Hoàng Nguyễn** and Manuel Clavel.
    Proving Correctness for SQL Implementations of OCL Constraints.
    Draft version, arXiv, 2024.

15. **Hoàng Nguyễn** and Manuel Clavel.
    Optimising Fine-Grained Access Control Policy Enforcement for Database Queries.
    Draft version, arXiv, 2022.

Artifacts and Tools

16. **Hoàng Nguyễn**, Srđan Krstić, and David Basin.
    MDP: Model-driven Privacy tools.
    https://nuactiongui.github.io

17. **Hoàng Nguyễn** and Manuel Clavel.
    OCL2PSQL Mapping.
    https://github.com/MoDELSVGU/OCL2PSQL

18. **Hoàng Nguyễn** and Manuel Clavel.

SQLSI: SQL Security Injector.
https://github.com/MoDELSVGU/SQLSI

19. **Hoàng Nguyễn** and Manuel Clavel.
SQL2MSFOL: SQL statements to Many-Sorted First-Order Logic.
https://github.com/MoDELSVGU/SQL2MSFOL

Teaching Activities

Real-Time Systems — Spring 2027 (upcoming)
Bachelor's Course in Computer Science, Vietnamese-German University — Vietnam
– Lecturer (Spring 2027)

IT Security — Fall 2026
Bachelor's Course in Computer Science, Vietnamese-German University — Vietnam
– Lab Assistant (Fall 2026)

Information Security — Spring 2023 – Spring 2025
Bachelor's Course in Computer Science, ETH Zurich — Switzerland
– Teaching Assistant (Spring 2023, Spring 2024, Spring 2025)

Security Engineering — Fall 2022 – Fall 2025
Master's Course in Computer Science, ETH Zurich — Switzerland
– Head of the Teaching Assistants (Fall 2023, Fall 2024)
– Teaching Assistant (Fall 2022, Fall 2025)

Supervision Activities

Misconception-Aware Interactive Python Learning for Beginners — Ongoing
Ms. Lê Minh Bảo Ngọc, Bachelor's Thesis, Vietnamese-German University — Vietnam
– Co-supervised with Assoc. Prof. Dr. Manuel Clavel

Misconception-Aware Interactive Python Learning for Beginners — Completed
Mr. Dương Võ Thiên Bảo, Bachelor's Thesis, Vietnamese-German University — Vietnam
– Co-supervised with Assoc. Prof. Dr. Manuel Clavel and Dr. Lê Lam Sơn

Formal Reasoning about Data Protection — Completed
Mr. Andrei Cotor, Master's Thesis, ETH Zurich — Switzerland
– Co-supervised with Assoc. Prof. Dr. Srđan Krstić

νActionGUI: Process Permissions and More Data Subject Rights — Completed
Mr. Andrei Cotor, Master's Semester Project, ETH Zurich — Switzerland
– Co-supervised with Assoc. Prof. Dr. Srđan Krstić

Privacy-Aware Object-Relational Mappers — Completed
Ms. Sofie Daniels, Master's Thesis, ETH Zurich — Switzerland
– Co-supervised with Assoc. Prof. Dr. Srđan Krstić

νActionGUI — Completed
Mr. Raymond Nasr, Master's Semester Project, ETH Zurich — Switzerland
– Co-supervised with Assoc. Prof. Dr. Srđan Krstić

Automated Threat Modeling for Industrial Control Systems — Completed
Mr. Mehdi Berrada, Master's Semester Project, ETH Zurich — Switzerland

Model Transformations for Privacy-by-Design Completed
Mr. Daniel Gradwohl, Bachelor's Thesis, ETH Zurich Switzerland
– Co-supervised with Assoc. Prof. Dr. Srđan Krstić

Secure Object-Relational Mappers Completed
Mr. Karlo Piskor, Bachelor's Thesis, ETH Zurich Switzerland
– Co-supervised with Assoc. Prof. Dr. Srđan Krstić

Fine-Grained Access Control in NoSQL Completed
Ms. Lê Minh Thư, Bachelor's Thesis, Vietnamese-German University Vietnam
– Co-supervised with Assoc. Prof. Dr. Manuel Clavel

A Web-Based Validator for Data Constraints Completed
Ms. Phạm Hồ Linh Thư, Bachelor's Thesis, Vietnamese-German University Vietnam
– Co-supervised with Assoc. Prof. Dr. Manuel Clavel

A Web-Based Intelligent Editor for Security Modeling Completed
Mr. Hoàng Thái Dương, Bachelor's Thesis, Vietnamese-German University Vietnam
– Co-supervised with Assoc. Prof. Dr. Manuel Clavel

Intelligent Web-Based Editor for Data Modeling Completed
Mr. Lữ Đình Tuấn, Bachelor's Thesis, Vietnamese-German University Vietnam
– Co-supervised with Assoc. Prof. Dr. Manuel Clavel

Voluntary Activities

Master Trainer Sep 2026 – Present
ASEAN Foundation, SCAM Ready ASEAN Vietnam
– Training communities to identify and prevent online scams

Mentor Apr 2024 – Dec 2024
Vietnamese-German University, Student-Alumni Mentoring Program Vietnam
– Mentoring computer science undergraduates

Translator Feb 2019 – Apr 2019
Như Chưa Hề Có Cuộc Chia Ly Vietnam
– Volunteer translation of program materials to reach international audiences